SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence COURSE CODE: CSA1709

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS:2 Total Marks:20

<u>Descriptive Test</u>

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Code of Conduct:

I _____k.abhishek dass / 192212301_______________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	5
Explanation	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	5
Application	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	4
Organization	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	3
Accuracy	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	2
Clarity	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	1

ANSWERS

1. Types of Intelligent Agents & Functionality in Smart Homes

A. Simple Reflex Agent

- **Functionality in Smart Home:**
  - **Lighting:** *IF motion detected in hallway THEN turn on lights.*
  - **Security:** *IF door glass breaks THEN trigger local alarm.*
- **Limitations:** Fails in partially observable or dynamic environments (e.g., keeping lights on while a user sits still reading).
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B. Model-Based Reflex Agent

- **Mechanism:** Maintains an **internal state** (a model of the world) to track unobserved aspects of the current state and account for past history.
- **Functionality in Smart Home:**
  - **Temperature/HVAC:** It remembers that the AC was turned on 10 minutes ago, allowing it to predict that temperature will drop soon without constantly toggling the compressor.
  - **Security:** Tracks whether the homeowner actively armed the security system before triggering an intrusion protocol when motion is detected.

C. Goal-Based Agent

- **Mechanism:** Combines state information with explicit **goal descriptions** to evaluate which sequence of actions will achieve the desired outcome.
- **Functionality in Smart Home:**
  - **Energy & Comfort:** Goal: *"Maintain indoor temperature at 22°C by 6:00 PM."* The agent plans ahead to start precooling/preheating at 5:15 PM based on current thermal decay rates.

D. Utility-Based Agent

- **Mechanism:** Uses a **utility function** to map states to a real number measuring "happiness" or trade-offs when goals conflict.
- **Functionality in Smart Home:**

2. Justification & Recommended Architecture

A single agent type is insufficient:

- A **Simple Reflex Agent** is too rigid to learn user preferences or manage trade-offs.
- A pure **Utility-Based Agent** introduces unnecessary computational overhead for safety-critical tasks.

The Ideal Solution: A Hybrid Learning Agent Architecture

1. **Reflex Subsystem (Safety/Security):** Uses simple condition-action rules for immediate responses (e.g., triggering gas leak shutoffs or intrusion alarms) without latency.

2. **Utility-Based Core (Comfort & Energy):** Manages climate control and routine lighting by evaluating trade-offs between dynamic electricity prices and user comfort ratings.
3. **Learning Component:** Observes daily user manual overrides (e.g., lowering temperature at night) to adapt utility weights and learn preferences automatically over time.

2.A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice

1. Application of Search Algorithms

Uniform Cost Search (UCS)

- **How it works:** Expands the node with the lowest path cost $g(n)$ using a priority queue.
- **Maze Application:** Ideal if maze actions have variable costs (e.g., moving straight costs 1 unit, turning costs 2, moving through rough terrain costs 3). UCS guarantees finding the cheapest path to the exit.

Iterative Deepening Search (IDS)

- **How it works:** Combines Depth-First Search (DFS) space efficiency with Breadth-First Search (BFS) completeness by iteratively increasing the depth limit ($d = 0,1,2, \ldots$).
- **Maze Application:** Ideal when step costs are uniform (e.g., each move = 1 step). It explores all paths up to depth d before moving deeper.

2. Complexity Analysis

Let b be the branching factor (number of available moves, e.g., $b \le 4$), d be the depth of the shallowest goal, and C^* be the cost of the optimal solution (with minimum edge cost ϵ).

Metric	Uniform Cost Search (UCS)	Iterative Deepening Search (IDS)
Time Complexity	$\mathcal{O}\left(b^{1+\lfloor C^*/\epsilon \rfloor}\right)$	$\mathcal{O}(b^d)$
Space Complexity	$\mathcal{O}\left(b^{1+\lfloor C^*/\epsilon \rfloor}\right)$ *(Stores all nodes in memory)*	$\mathcal{O}(b \cdot d)$ *(Memory-efficient)*
Completeness	**Yes** (if edge costs are positive: $c \ge \epsilon > 0$)	**Yes** (if b is finite)
Optimality	**Optimal** for arbitrary non-negative step costs	**Optimal** if all step costs are equal

- **UCS Advantage:** Guarantees optimal path cost under non-uniform move costs.
- **UCS Disadvantage:** High memory footprint (can run out of RAM in large mazes).
- **IDS Advantage:** Extremely low memory consumption ($\mathcal{O}(bd)$) while retaining BFS optimality for unit costs.

- **IDS Disadvantage:** Re-evaluates upper-level nodes multiple times across iterations.

3. Agent Design & Decision Making

Agent structure dictates how search algorithms are executed:

1. **Problem-Solving Agent (Goal-Based Architecture):** Formulates a sequence of atomic actions *before* execution. In an **unknown** environment, this leads to an **Online Search Problem**, where the agent alternates between sensing, building a local map, and searching.
2. **Model-Based Refinement:** The agent maintains a state map of explored vs. unexplored maze coordinates to avoid infinite looping and re-searching known dead ends.

4. Recommended Strategy & Justification

Best Combination: A Model-Based Online Search Agent paired with Iterative Deepening Search (IDS)

- **Justification:**
    1. **Memory Efficiency in Physical Hardware:** Autonomous maze-navigating robots typically operate with limited onboard memory. IDS requires only $O(bd)$ space, preventing memory overflow.
    2. **Uniform Movement Cost:** In grid-based mazes, standard steps (up, down, left, right) have uniform costs, making IDS optimal in both path length and step cost without needing the overhead of UCS.
    3. **Online Mapping Adaptability:** Because the maze is unknown, the robot must incrementally map the space. IDS allows low-cost depth-bound exploration cycles as new corridors are discovered.